System Validation Report

Evidence generated: 30 October 2026, 23:31 IST. Baseline commit: 9ea6b597a8d51b4070b80182b056575b8c183aeb. Tests include the supplied working-tree changes.

This is an automated local verification record. It is not UAT sign-off or deployment certification. Simulated business dates in November are test inputs; execution dates are not backdated.

Executed scope

External HTTP API/security acceptance: **50 passed / 0 failed / 0 skipped / 50 total** across 9 suites. Backend unit/integration/TLS combined: **265 passed / 0 failed / 0 skipped / 265 total** across 23 suites. The frontend production build is a compilation check, not a browser test.

MongoDB is a real temporary local database process. Gates, payment confirmations and notifications use project simulators, signed test webhooks and a notification outbox. Reset hooks are enabled only in test mode.

Local performance observation

The supplementary benchmark used 20 concurrent gate lanes, three cycles per lane, 1000 historical stays and 600 slots. It is a short single-instance local run, not the deployed 100/1000-user JMeter experiment.

Step	Samples	p50 ms	p90 ms	p99 ms
Entry	60	1454.1	4376.7	4617.8
Payment	60	833.9	1265.5	1295.6
Webhook	60	334.3	372.2	567.0
Exit	60	1284.3	1572.0	1809.1

Observed failed cycles: 0. Raw evidence: evidence/gate-benchmark.log.

The local response-time budget was missed. Zero failed cycles does not mean the latency requirement passed. This short local measurement is sensitive to machine contention and does not establish deployed capacity, but it must be investigated rather than replaced with a better sample. The benchmark program reports measurements and returns success for execution; its process exit code is not a performance-budget assertion.

Requirement traceability

Case	Requirement	Outcome
TC-Entry-01	VPS-F-001	PASS
TC-Entry-02	VPS-F-002	PASS
TC-Entry-03	VPS-F-003	PASS
TC-Entry-04	VPS-F-004	PASS
TC-Exit-01	VPS-F-010	PASS
TC-Exit-02	VPS-F-011	PASS
TC-Exit-03	VPS-F-012	PASS
TC-Exit-04	VPS-F-013	PASS
TC-Res-01	VPS-F-020	PASS
TC-Res-02	VPS-F-021	PASS
TC-Res-03	VPS-F-022	PASS
TC-Res-04	VPS-F-023	PASS
TC-Pay-01	VPS-F-030	PASS
TC-Pay-02	VPS-F-031	PASS
TC-Pay-03	VPS-F-032	PASS
TC-Admin-01	VPS-F-040	PASS
TC-Admin-02	VPS-F-041	PASS
TC-Admin-03	VPS-F-042	PASS
TC-Admin-04	VPS-F-043	PASS
TC-Sec-01	VPS-NF-003	PASS
TC-Sec-02	VPS-SR-001	PARTIAL - local evidence only
TC-Sec-03	VPS-SR-002	PASS
TC-Sec-04	VPS-SR-003	PASS
TC-Sec-05	VPS-SR-004	PASS
TC-Sec-06	VPS-SR-005	PASS
TC-OPS-01	VPS-NF-004	PASS (synthetic integration)
TC-Perf-01	VPS-NF-001	FAIL - local response-time sample
TC-Perf-02	VPS-NF-005	NOT RUN
TC-UAT-01	STP-5	NOT RUN
TC-UAT-02	STP-5	NOT RUN
TC-INT-01	STP-6	NOT RUN

Case	Requirement	Outcome
TC-OPS-02	VPS-NF-002	NOT RUN

Remaining Test Plan activities

Activity	Status	Reason / scope
TC-Perf-02 / VPS-NF-005	NOT RUN	Deployed 100-to-1000-user JMeter experiment and representative horizontal topology unavailable.
TC-Perf-01 / VPS-NF-001	FAIL (local sample)	Local entry/exit p90 exceeded 3000 ms; deployed baseline still unverified.
TC-Sec-02 / VPS-SR-001	PARTIAL	Local TLS/config suite only; external testssl.sh and OWASP ZAP scan not executed.
TC-OPS-02 / VPS-NF-002	NOT RUN	A full month of uptime observations is not available.
TC-UAT-01 and TC-UAT-02	NOT RUN	Real-user acceptance, browser workflows, screen-reader, contrast and print review require execution.
TC-INT-01	NOT RUN	Hardware, external gateway sandbox and actual email/SMS delivery not verified; current hooks use simulators/outbox.
TC-Sec-01 / compliance	PARTIAL	Automated no-secret/PAN checks are evidence, not a regulatory compliance assessment.

Recorded metrics

Measure	Observed value
Acceptance assertion execution	50 / 50
Acceptance pass rate	100.0% of recorded assertions
Backend assertion execution	265 / 265
Catalog case groups passing locally	25 / 32; partial and not-run groups retained
Defect density / aging	Not computed: no consistent historical defect population or lifecycle dates supplied

Exit decision

Not ready for unconditional Test Plan sign-off. Passing automated functional checks does not satisfy the plan requirement that every test case pass. Complete the pending external, performance, accessibility, UAT and monitoring checks before an authorized reviewer signs. No approval or signature is fabricated.